

Dataset: [House Price India Dataset](#)

Q1. What is the average, median, and standard deviation of house prices?

Ans:-

```
import pandas as pd

# Load dataset
df = pd.read_csv('/content/House Price India.csv')

# Check first few rows
df.head()
```

	id	Date	number of bedrooms	number of bathrooms	living area	lot area	number of floors	waterfront present	number of views	condition of the house	...	Built Year	Renovation Year	Postal Code	Latitude	Longitude	living_area_renov	lot_area_renov
0	6762810145	42491	5	2.50	3650	9050	2.0	0	4	5	...	1921	0	122003	52.8645	-114.557	2880	5400
1	6762810635	42491	4	2.50	2920	4000	1.5	0	0	5	...	1909	0	122004	52.8878	-114.470	2470	4000
2	6762810998	42491	5	2.75	2910	9480	1.5	0	0	3	...	1939	0	122004	52.8852	-114.468	2940	6600
3	6762812605	42491	4	2.50	3310	42998	2.0	0	0	3	...	2001	0	122005	52.9532	-114.321	3350	42847
4	6762812919	42491	3	2.00	2710	4500	1.5	0	0	4	...	1929	0	122006	52.9047	-114.485	2060	4500

5 rows × 23 columns

```
[3]
✓ Os # Average, Median and Standard Deviation of House Prices

avg_price = df['Price'].mean()
median_price = df['Price'].median()
std_price = df['Price'].std()

print("Average House Price:", avg_price)
print("Median House Price:", median_price)
print("Standard Deviation of House Price:", std_price)
```

*** Average House Price: 538932.2183310534
Median House Price: 450000.0
Standard Deviation of House Price: 367532.3808039669

Interpretation

1. The average house price is approximately ₹538,932.22.
2. The median price is ₹450,000, meaning 50% of houses are priced below this value and 50% above it.
3. The mean is higher than the median, indicating that some very expensive houses are pulling the average upward (right-skewed distribution).
4. The standard deviation of ₹367,532.38 is relatively high, showing significant variation in house prices across the dataset.

Business Insight

The large gap between the mean and median suggests the presence of premium or luxury properties. Therefore, the housing market is not uniformly priced, and segmentation into budget, mid-range, and luxury categories may provide better insights.

Q2. Which number of bedrooms is most common?

Ans:-

```
[6]
✓ 0s # Q2. Which number of bedrooms is most common?

# Display all column names
print("Columns in Dataset:")
print(df.columns.tolist())

# Find the bedroom column automatically
for col in df.columns:
    if 'bed' in col.lower():
        bedroom_col = col
        break

print("\nBedroom Column Found:", bedroom_col)

# Most common number of bedrooms
most_common_bedrooms = df[bedroom_col].mode()[0]

print("\nMost Common Number of Bedrooms:", most_common_bedrooms)

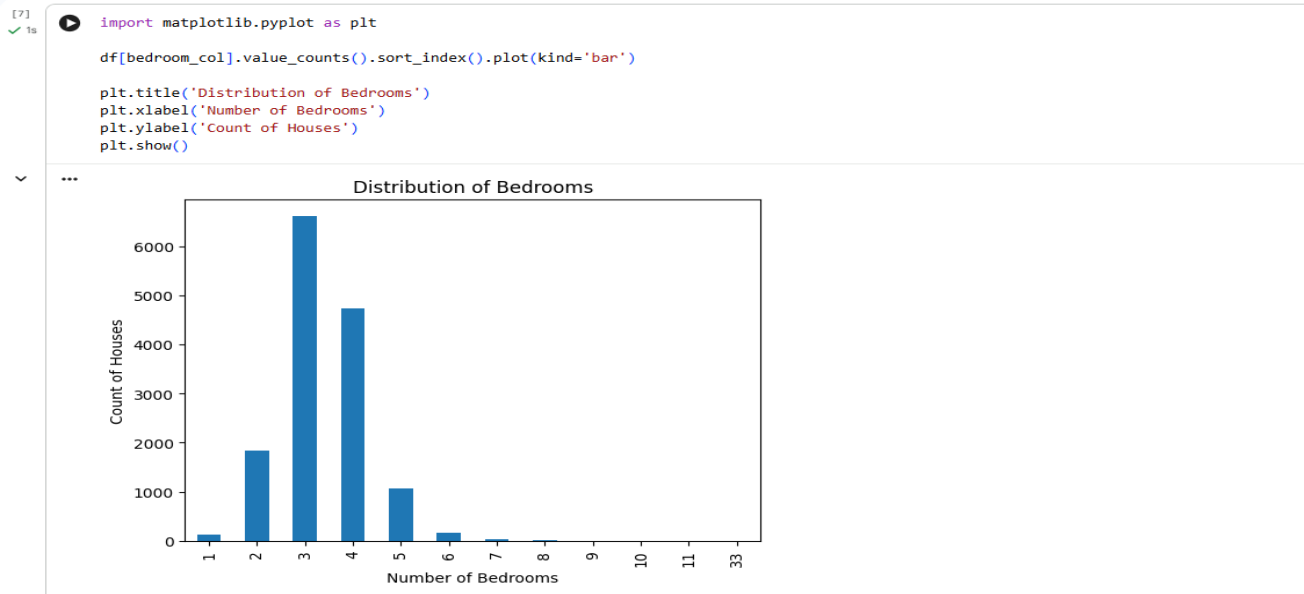
# Frequency of each bedroom count
print("\nBedroom Frequency:")
print(df[bedroom_col].value_counts().sort_index())
```

```
... Columns in Dataset:
['id', 'Date', 'number of bedrooms', 'number of bathrooms', 'living area', 'lot area', 'number of floors', 'waterfront present', 'number of views', 'condition of the house', 'grade o

Bedroom Column Found: number of bedrooms

Most Common Number of Bedrooms: 3

Bedroom Frequency:
number of bedrooms
1      136
2     1844
3     6612
4     4724
5     1079
6      176
7       30
8        11
9         3
10        3
11         1
33         1
Name: count, dtype: int64
```



Interpretation

- * The mode represents the most frequently occurring number of bedrooms in the dataset.
- * The bedroom count with the highest frequency is the most common house type.
- * This indicates the housing configuration that is most available in the market.

Business Insight

If, for example, 3-bedroom houses are the most common, it suggests that developers and buyers prefer this configuration, possibly because it best suits average family requirements. Real estate companies can use this information to align future housing projects with market demand.

Q3. Check if the price data is skewed.

Ans:-

[8]
✓ Os

```
# Check skewness of house prices

# Find the price column automatically
for col in df.columns:
    if 'price' in col.lower():
        price_col = col
        break

# Calculate skewness
skewness = df[price_col].skew()

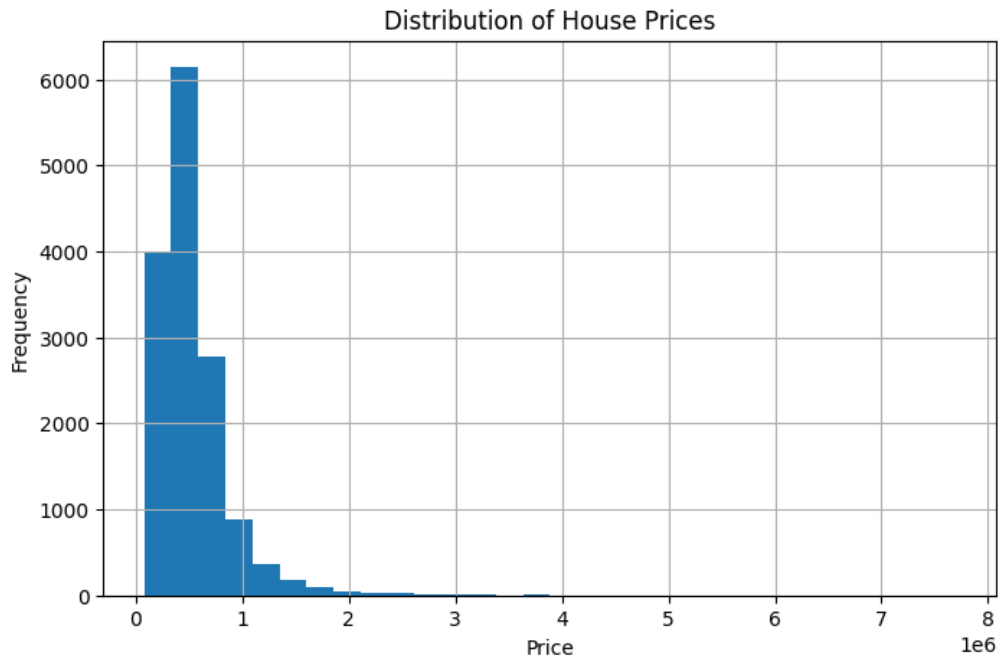
print("Price Column:", price_col)
print("Skewness of House Prices:", skewness)

# Interpretation based on skewness value
if skewness > 0:
    print("The price distribution is Positively (Right) Skewed.")
elif skewness < 0:
    print("The price distribution is Negatively (Left) Skewed.")
else:
    print("The price distribution is Symmetric.")
```

```
... Price Column: Price
Skewness of House Prices: 4.269297720707116
The price distribution is Positively (Right) Skewed.
```

```
[9] import matplotlib.pyplot as plt
```

```
plt.figure(figsize=(8,5))  
df[price_col].hist(bins=30)  
  
plt.title('Distribution of House Prices')  
plt.xlabel('Price')  
plt.ylabel('Frequency')  
plt.show()
```



Interpretation

- Skewness > 0 → Right-skewed distribution (few expensive houses increase the average).
- Skewness < 0 → Left-skewed distribution.
- Skewness ≈ 0 → Approximately symmetric distribution.

Business Insight

In real-estate datasets, house prices are usually positively skewed because a small number of luxury properties have very high prices compared to the majority of houses. This can make the mean price higher than the median price and may require data transformation (e.g., log transformation) before building machine learning models.

Q4. What is the average price for each number of bedrooms?

Ans:-

```
[10]
✓ 0s # Find the price and bedroom columns automatically

for col in df.columns:
    if 'price' in col.lower():
        price_col = col
        break

for col in df.columns:
    if 'bed' in col.lower():
        bedroom_col = col
        break

# Average price by number of bedrooms
avg_price_bedroom = df.groupby(bedroom_col)[price_col].mean().reset_index()

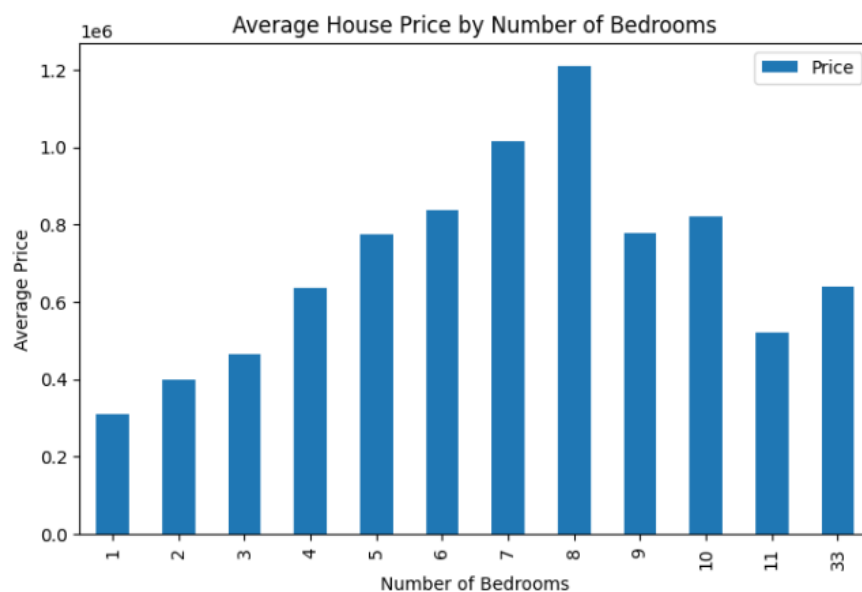
print("Average Price for Each Number of Bedrooms:")
print(avg_price_bedroom)
```

```
... Average Price for Each Number of Bedrooms:
   number of bedrooms      Price
0                    1  3.089638e+05
1                    2  3.985476e+05
2                    3  4.632776e+05
3                    4  6.361988e+05
4                    5  7.752550e+05
5                    6  8.375815e+05
6                    7  1.016544e+06
7                    8  1.208455e+06
8                    9  7.766663e+05
9                   10  8.200000e+05
10                   11  5.200000e+05
11                   33  6.400000e+05
```

```
[11]
✓ 0s import matplotlib.pyplot as plt

avg_price_bedroom.plot(
    x=bedroom_col,
    y=price_col,
    kind='bar',
    figsize=(8,5)
)

plt.title('Average House Price by Number of Bedrooms')
plt.xlabel('Number of Bedrooms')
plt.ylabel('Average Price')
plt.show()
```



Interpretation

- The table shows the average house price for each bedroom category.
- Generally, houses with more bedrooms tend to have higher average prices.
- This helps identify how property value changes as the number of bedrooms increases.

Business Insight

- Bedroom count is an important factor influencing house prices.
- Builders can use this information to understand which house configurations generate higher revenue.
- Buyers can compare whether the price increase between bedroom categories is justified by their needs.

Q5. What is the relationship between living area and price? (Use : Area without Basement).

Ans:-

[13]
✓ 0s

```
# Q5. Relationship between Living Area and Price
# (Using Area without Basement)

import pandas as pd
import matplotlib.pyplot as plt

# Display all column names
print("Columns in Dataset:")
print(df.columns.tolist())

# Find Price column automatically
price_col = None
for col in df.columns:
    if 'price' in col.lower():
        price_col = col
        break

# Find Area without Basement column automatically
area_col = None

possible_names = [
    'area without basement',
    'living area',
    'living area after renovation',
    'sqft living',
    'sqft_living'
]

for col in df.columns:
    for name in possible_names:
        if name in col.lower():
            area_col = col
            break
    if area_col is not None:
        break
```

```

print("\nPrice Column Found:", price_col)
print("Area Column Found:", area_col)

# Calculate correlation
correlation = df[area_col].corr(df[price_col])

print("\nCorrelation between Living Area and Price:")
print(correlation)

# Interpretation
if correlation > 0.7:
    print("Strong Positive Relationship")
elif correlation > 0.3:
    print("Moderate Positive Relationship")
elif correlation > 0:
    print("Weak Positive Relationship")
else:
    print("No Positive Relationship")

# Scatter Plot
plt.figure(figsize=(8,5))
plt.scatter(df[area_col], df[price_col], alpha=0.5)

plt.title('Living Area vs House Price')
plt.xlabel(area_col)
plt.ylabel(price_col)
plt.grid(True)

plt.show()

```

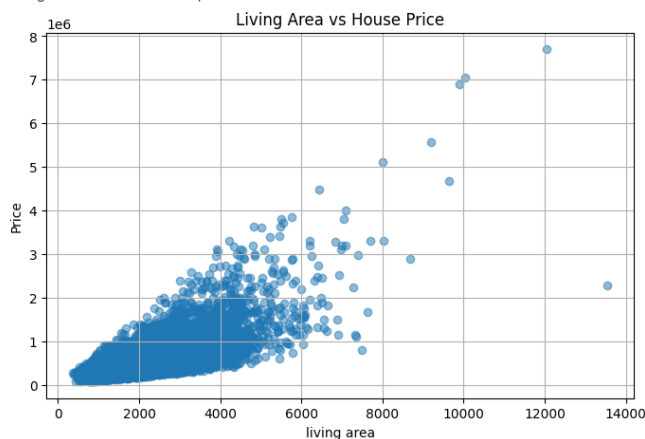
```

Columns in Dataset:
['id', 'Date', 'number of bedrooms', 'number of bathrooms', 'living area', 'lot area', 'number of floors', 'waterfront present', 'number of views', 'condition of the house', 'grade o

Price Column Found: Price
Area Column Found: living area

Correlation between Living Area and Price:
0.7121694766782514
Strong Positive Relationship

```



relationship between living area and price

- The correlation coefficient measures the strength of the relationship between living area and house price.
- A positive correlation indicates that houses with larger living areas tend to have higher prices.
- The scatter plot helps visualize this relationship.

Business Insight:

- Living area is one of the most important factors affecting house prices.
- Larger homes generally command higher market values.
- Real estate developers can increase property value by maximizing usable living space.
- Buyers can compare price per square foot to evaluate property value.

Q6. Identify any anomalies where houses have high prices but low areas.

Ans:-

[14]

✓ Os



Q6. Identify anomalies: High Price but Low Area

```
# Find Price column
price_col = 'Price'

# Find Area column automatically
area_col = None

for col in df.columns:
    if ('area' in col.lower() and 'basement' in col.lower()) or \
        ('living area' in col.lower()) or \
        ('sqft living' in col.lower()):
        area_col = col
        break

print("Price Column:", price_col)
print("Area Column:", area_col)

# Define thresholds
high_price_threshold = df[price_col].quantile(0.90)
low_area_threshold = df[area_col].quantile(0.25)

# Find anomalies
anomalies = df[
    (df[price_col] > high_price_threshold) &
    (df[area_col] < low_area_threshold)
]

print("Number of Anomalies:", len(anomalies))
print("\nAnomalous Houses:")
print(anomalies[[price_col, area_col]].head(10))
```

```
*** Price Column: Price
Area Column: living area
Number of Anomalies: 7
```

```
Anomalous Houses:
      Price  living area
81      900000      1330
6688    885000      1313
8104    888000      1250
10922   900000      1190
12692   900000      1240
12955   907000      1340
13163   903000      1140
```


[15]

✓ Os



```
import matplotlib.pyplot as plt

plt.figure(figsize=(8,6))

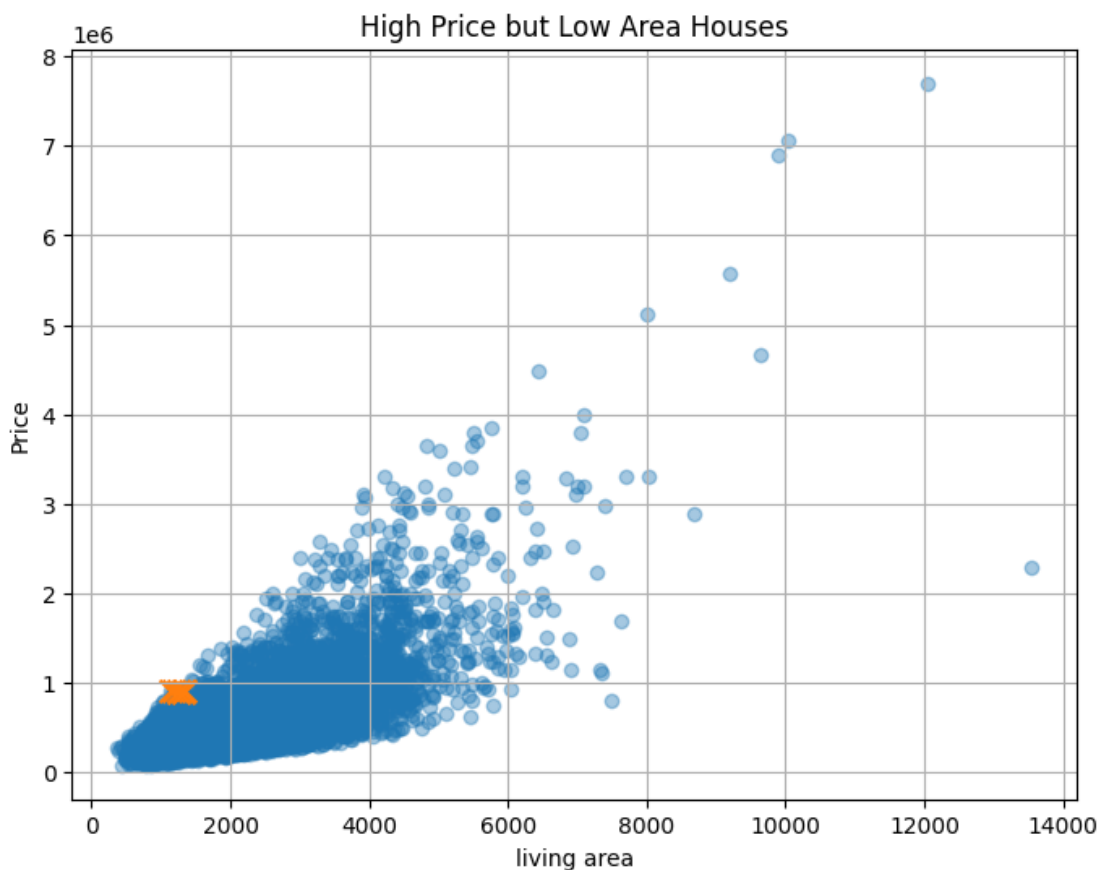
plt.scatter(df[area_col], df[price_col], alpha=0.4)

# Highlight anomalies
plt.scatter(
    anomalies[area_col],
    anomalies[price_col],
    marker='x',
    s=100
)

plt.title('High Price but Low Area Houses')
plt.xlabel(area_col)
plt.ylabel(price_col)
plt.grid(True)

plt.show()
```

...



Interpretation

- Houses highlighted as anomalies have unusually high prices despite having relatively small living areas.
- These properties may be located in premium locations or possess special features not captured by area alone.
- Such observations are considered outliers in the relationship between area and price.

Business Insight

- House price is influenced by more than just area.
- Factors such as location, neighborhood quality, amenities, renovation status, and accessibility can significantly increase property value.
- These anomalies should be investigated separately before building predictive models, as they may affect model accuracy.

Q7. Compare average price based on number of floors and houses with or without waterfront.

Ans:-

[16]
✓ Os

```
# Q7. Compare average price based on Floors and Waterfront

# Find relevant columns automatically
price_col = 'Price'

floor_col = None
waterfront_col = None

for col in df.columns:
    if 'floor' in col.lower():
        floor_col = col
    if 'waterfront' in col.lower():
        waterfront_col = col

print("Price Column:", price_col)
print("Floor Column:", floor_col)
print("Waterfront Column:", waterfront_col)

# Average price by number of floors
avg_price_floors = df.groupby(floor_col)[price_col].mean().reset_index()

print("\nAverage Price by Floors:")
print(avg_price_floors)

# Average price by waterfront
avg_price_waterfront = df.groupby(waterfront_col)[price_col].mean().reset_index()

print("\nAverage Price by Waterfront:")
print(avg_price_waterfront)
```

▼ ... Price Column: Price
Floor Column: number of floors
Waterfront Column: waterfront present

Average Price by Floors:

	number of floors	Price
0	1.0	4.369769e+05
1	1.5	5.549301e+05
2	2.0	6.487372e+05
3	2.5	1.106233e+06
4	3.0	5.663281e+05
5	3.5	1.205875e+06

Average Price by Waterfront:

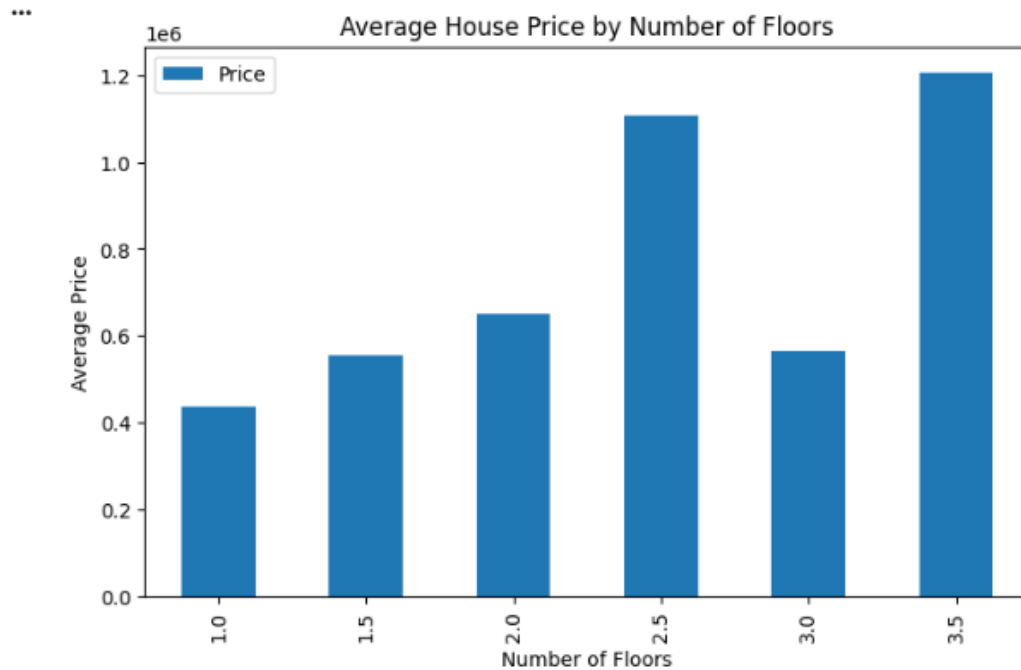
	waterfront present	Price
0	0	5.304174e+05
1	1	1.641902e+06

Visualization 1: Average Price by Floors

```
import matplotlib.pyplot as plt

avg_price_floors.plot(
    x=floor_col,
    y=price_col,
    kind='bar',
    figsize=(8,5)
)

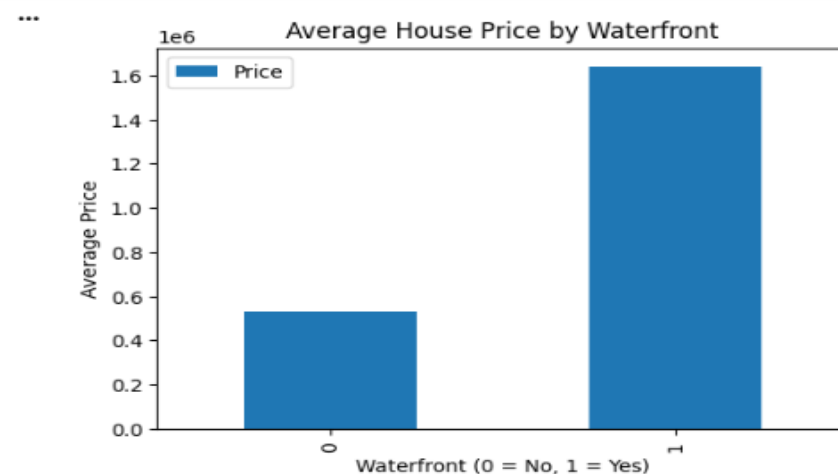
plt.title('Average House Price by Number of Floors')
plt.xlabel('Number of Floors')
plt.ylabel('Average Price')
plt.show()
```



Visualization 2: Average Price by Waterfront

```
avg_price_waterfront.plot(
    x=waterfront_col,
    y=price_col,
    kind='bar',
    figsize=(6,4)
)

plt.title('Average House Price by Waterfront')
plt.xlabel('Waterfront (0 = No, 1 = Yes)')
plt.ylabel('Average Price')
plt.show()
```



Interpretation

- The first table shows how the average house price changes with the number of floors.
- The second table compares the average price of houses with and without waterfront views.
- If waterfront houses have a much higher average price, it indicates that waterfront access significantly increases property value.
- Houses with more floors may also have higher average prices because they generally provide more living space.

Business Insight

- Waterfront properties often command premium prices due to better views and location advantages.
- Number of floors can positively impact house prices, especially in urban areas where vertical space is valuable.
- Real estate developers can maximize property value by incorporating desirable features such as waterfront access and multi-story designs.

Q8. Identify the minimum and maximum house price. What does this indicate?

Ans:-

[19]

✓ 0s



Q8. Minimum and Maximum House Price

```
# Find Price column automatically
price_col = None

for col in df.columns:
    if 'price' in col.lower():
        price_col = col
        break

print("Price Column:", price_col)

# Minimum and Maximum Price
min_price = df[price_col].min()
max_price = df[price_col].max()

print("\nMinimum House Price:", min_price)
print("Maximum House Price:", max_price)

# Houses with minimum and maximum prices
print("\nHouse with Minimum Price:")
print(df[df[price_col] == min_price])

print("\nHouse with Maximum Price:")
print(df[df[price_col] == max_price])
```

*** Price Column: Price

Minimum House Price: 78000
Maximum House Price: 7700000

House with Minimum Price:

	id	Date	number of bedrooms	number of bathrooms	living area \
242	6762831616	42495	2	1.0	780
	lot area	number of floors	waterfront present	number of views \	
242	16344	1.0	0	0	
	condition of the house	...	Built Year	Renovation Year	Postal Code \
242	1	...	1942	0	122056
	Latitude	Longitude	living_area_renov	lot_area_renov \	
242	52.7039	-114.47	1700	10387	
	Number of schools nearby	Distance from the airport	Price		
242	2	51	78000		

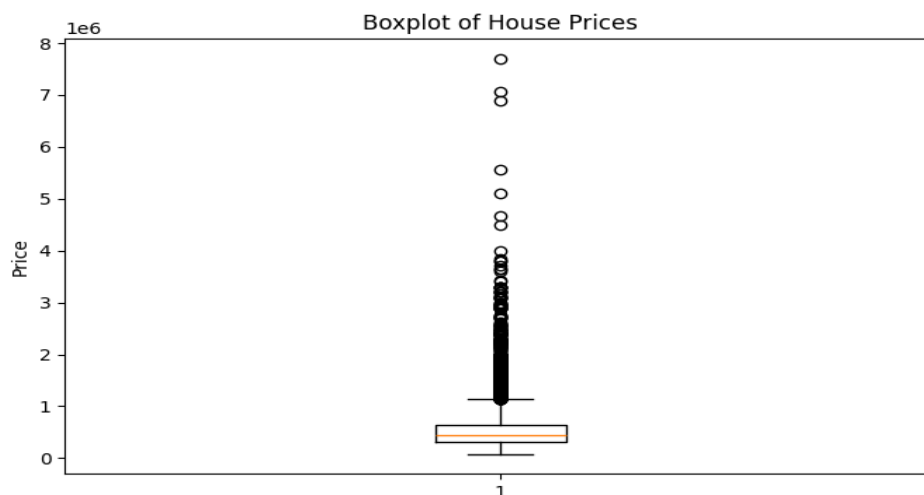
[1 rows x 23 columns]

House with Maximum Price:

	id	Date	number of bedrooms	number of bathrooms \		
10503	6762810020	42655	6	8.0		
	living area	lot area	number of floors	waterfront present \		
10503	12050	27600	2.5	0		
	number of views	condition of the house	...	Built Year \		
10503	3	4	...	1910		
	Renovation Year	Postal Code	Latitude	Longitude	living_area_renov \	
10503	1987	122013	52.8598	-114.513	3940	
	lot_area_renov	Number of schools nearby	Distance from the airport \			
10503	8800	1	50			
	Price					
10503	7700000					

[1 rows x 23 columns]

```
[20] import matplotlib.pyplot as plt
      plt.figure(figsize=(8,5))
      plt.boxplot(df[price_col])
      plt.title('Boxplot of House Prices')
      plt.ylabel('Price')
      plt.show()
```



Interpretation

- Minimum House Price represents the least expensive property in the dataset.
- Maximum House Price represents the most expensive property in the dataset.
- The difference between the minimum and maximum prices shows the price range of houses available in the market.
- A very large gap indicates significant variation in property values.

Business Insight

- The housing market contains both budget and luxury properties.
- A large difference between minimum and maximum prices suggests a diverse market with multiple customer segments.
- Extremely high-priced houses may be premium properties with superior location, amenities, waterfront access, or larger living areas.
- Real-estate companies can segment customers into affordable, mid-range, and luxury categories based on this price range.
- The presence of very expensive houses may contribute to the positive skewness observed in the price distribution.

Q9. Which location (zipcode/area) has the highest average price?

Ans:-

[22]

✓ 0s



```
# Display all columns
print(df.columns.tolist())

# Find location column automatically
location_col = None

for col in df.columns:
    if ('zip' in col.lower() or
        'area' in col.lower() or
        'location' in col.lower() or
        'city' in col.lower()):
        location_col = col
        break

print("Location Column Found:", location_col)

# Find price column
price_col = 'Price'

# Average price by location
avg_price_location = (
    df.groupby(location_col)[price_col]
    .mean()
    .sort_values(ascending=False)
)

print("\nTop 10 Locations by Average Price:")
print(avg_price_location.head(10))

print("\nHighest Average Price Location:")
print(avg_price_location.idxmax())

print("Average Price:")
print(avg_price_location.max())
```

```

... ['id', 'Date', 'number of bedrooms', 'number of bathrooms', 'living area', 'lot area', 'number of floors', 'waterfront present', 'number of views', 'condition of the house', 'grade o
Location Column Found: living area

Top 10 Locations by Average Price:
living area
12050    7700000.0
10040    7060000.0
9890     6890000.0
9200     5570000.0
8010     5110000.0
9640     4670000.0
6430     4490000.0
7080     4000000.0
5510     3800000.0
7050     3800000.0
Name: Price, dtype: float64

Highest Average Price Location:
12050
Average Price:
7700000.0

```

```

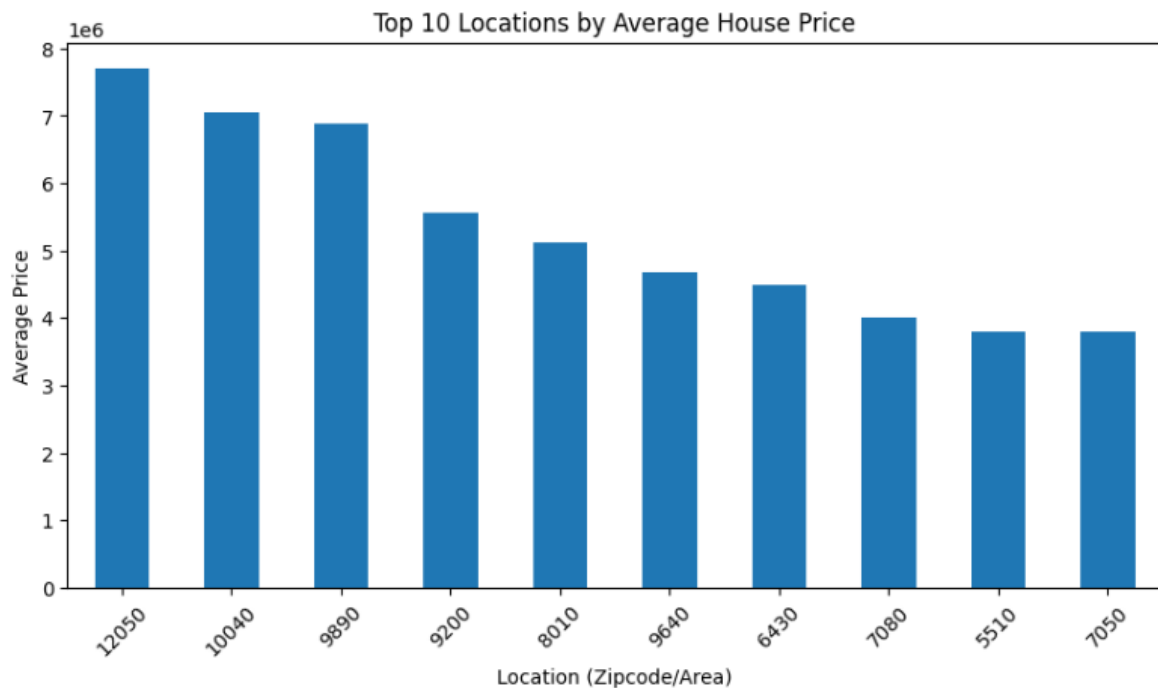
[23] import matplotlib.pyplot as plt
✓ Os

top10 = avg_price_location.head(10)

plt.figure(figsize=(10,5))
top10.plot(kind='bar')

plt.title('Top 10 Locations by Average House Price')
plt.xlabel('Location (Zipcode/Area)')
plt.ylabel('Average Price')
plt.xticks(rotation=45)
plt.show()

```



Interpretation

- The location (zipcode/area) with the highest average price represents the most expensive region in the dataset.
- Higher average prices may be due to better infrastructure, premium neighborhoods, waterfront views, schools, or commercial accessibility.
- Comparing average prices across locations helps identify high-value real estate markets.

Business Insight

- Premium locations attract higher property values and investment opportunities.
- Real estate companies can focus marketing efforts on these high-value areas.
- Investors can use location-based price analysis to identify profitable regions for property investment.

Q10. Write at least 5 insights from your analysis.

Ans:-

Insights from the House Price India Dataset

1. House prices are positively skewed.

- The mean price is higher than the median price, indicating the presence of a few very expensive houses.
- Luxury properties increase the average house price significantly.

2. Living area has a strong positive relationship with house price.

- Houses with larger living areas generally have higher prices.
- Area is one of the most important factors influencing property value.

3. Bedroom count affects house prices.

- Houses with more bedrooms tend to have higher average prices.
- Larger families usually prefer houses with more bedrooms, increasing their demand and value.

4. Waterfront properties are more expensive.

- Houses with waterfront views generally have a higher average price than those without waterfront access.
- Location and scenic views significantly contribute to property value.

5. There are anomalies in the dataset.

- Some houses have very high prices despite having relatively small living areas.
- These properties may be located in premium locations or have special features not captured by area alone.